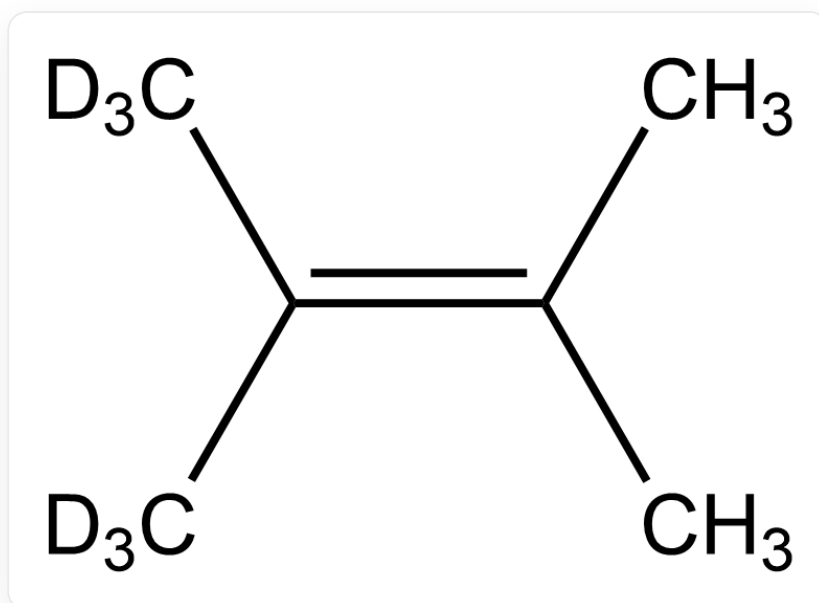


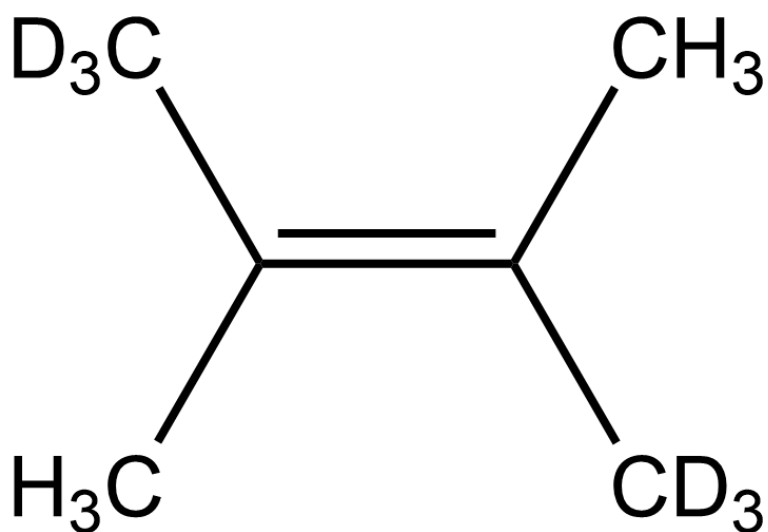
Question

Predict for which substrates the first-order kinetic isotope effect (KIE) can be observed when singlet oxygen $^1\text{O}_2$ reacts with them. Consider only the reaction with a single molecule.



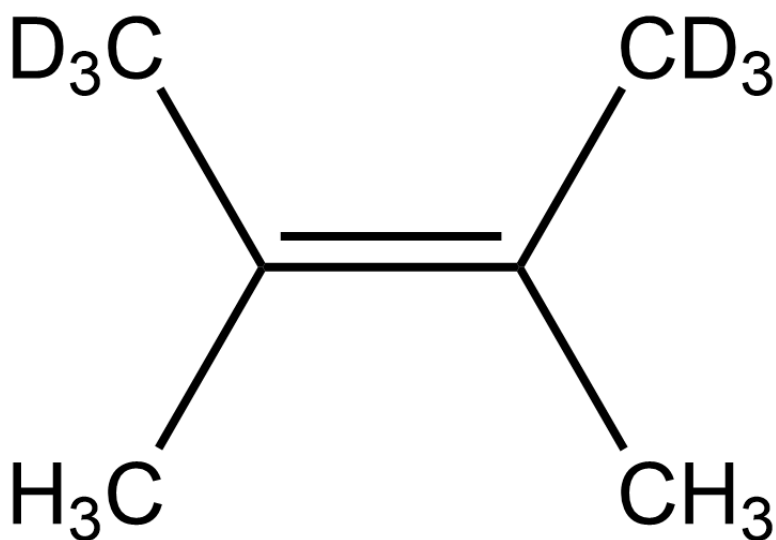
C/C(C)=C(C([2H])([2H])[2H])\C([2H])([2H])[2H], Substrate 1

Substrate 1



C/C(C([2H])([2H])[2H])=C(C)\C([2H])([2H])[2H], Substrate 2

Substrate 2



C/C(C([2H])([2H])[2H])=C(C)/C([2H])([2H])[2H], Substrate 3

Substrate 3

A. All other options are incorrect

B. Only option 1

C. Only option 2

D. Only option 3

E. Only options 1, 2

F. Only options 1, 3

G. Only options 2, 3

H. Option 1, 2, 3

Answer

Correct Answer: **E**

Detailed Explanation

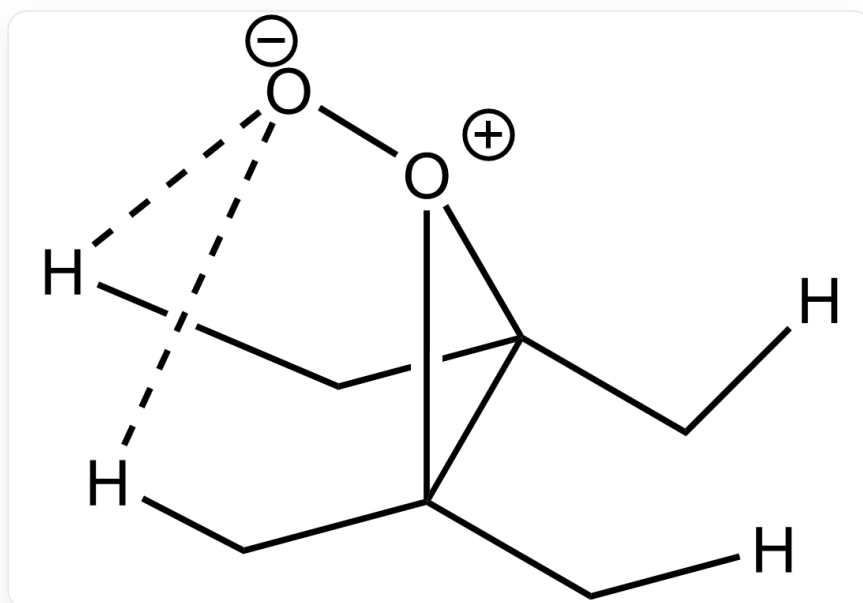
The kinetic isotope effect arises from the difference in zero-point energy between C-H and C-D bonds.

CHECKPOINT

1 PTS

The kinetic isotope effect arises from the difference in zero-point energy between C-H and C-D bonds

This is the transition state of the reaction



[H]CC(C[H])([O+][O-])C1(C[H])C[H]

For substrate 3, regardless of which side the peroxide faces in the intermediate, only one of H or D can be attacked, thus no KIE is generated.

CHECKPOINT

1 PTS

For substrate 3, regardless of which side the peroxide faces in the intermediate, only one of H or D can be attacked, thus no KIE is generated

Whereas for substrates 1 and 2, regardless of which side the peroxide faces in the intermediate, there is competition between attacking H and D, hence KIE exists.

CHECKPOINT

1 PTS

Whereas for substrates 1 and 2, regardless of which side the peroxide faces in the intermediate, there is competition between attacking H and D, hence KIE exists